

In Silico Design and Computational Validation of a Tri-Modular Genetic Circuit for Autonomous Detection and Bioremediation of Hexavalent Chromium in Closed-System Microcosms

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Abstract

Hexavalent chromium [Cr(VI)], a Group 1 IARC carcinogen, reaches up to 3.54 mg/L in Bangladesh’s Buriganga River (35-fold above the US EPA limit) due to unregulated tannery effluent. Existing remediation is passive, lacking real-time sensing, dose-responsive treatment, and self-termination. Furthermore, deploying engineered microorganisms in open rivers poses unquantified ecological risks.

Here, we present a fully computational design and validation of a tri-modular synthetic genetic circuit in a double-auxotrophic *Escherichia coli* chassis for closed-system microcosms. The orthogonal modules include: **(1)** a ChrB-sfGFP biosensor (pUC19) for nanomolar Cr(VI) detection; **(2)** a NemaA chromate reductase (pET-28a), driven by the identical P_{chrB} promoter, for dose-responsive Cr(VI) $\rightarrow$ Cr(III) reduction; **(3)** a dual-trigger holin-endolysin “Deadman” kill switch (pACYC184) enforcing programmed cell death via AND-gate logic, combining chromate depletion sensing with a 24-hour CI434-SsrA timer repressor.

Validation employs five methodologies: **(i)** 96-hour ODE modeling of module dynamics; **(ii)** comparative Michaelis-Menten modeling of a hypothetical NemaA^{*2+} catalytic variant, using kinetic parameters extrapolated from OYE-family precedent [15]; **(iii)** structural evaluation (PDB 8BPQ) via a contact-disruption score and geometric substrate placement, providing a *consistency check* for the NemaA^{*2+} hypothesis; **(iv)** a ribosome-allocation metabolic model demonstrating sustainability of the circuit’s $\sim 4.5\%$ proteome burden; and **(v)** Luria-Delbrück fluctuation analysis of the layered biocontainment. Under assumed parameters, this yields a 30-day viable escapee probability ($P_{\text{escape}} \approx 6 \times 10^{-17}$) in a 1,000-liter system, safely below the 10^{-15} threshold.

Sensitivity analysis reveals this safety margin is robust to a $10\times$ error in mutation rate or a $10\times$ reactor scale-up, but fails at a $100\times$ mutation rate error ($P_{\text{escape}} = 6 \times 10^{-15}$). Global Latin Hypercube Sampling ($N = 10,000$) across 8 parameters yields median $P_{\text{escape}} = 1.7 \times 10^{-17}$, but **20.6% of samples exceed the 10^{-15} threshold** under plausible joint-tail parameter combinations — indicating the containment margin is not yet robust and is contingent on empirical measurement of HGT transfer rates.

This system computationally achieves $>99\%$ reduction of $100\ \mu\text{M}$ Cr(VI) within 48 hours. Ultimately, this framework establishes a transparent *in silico* biosafety validation template for heavy-metal bioremediation, mathematically quantifying both performance and mutation-driven failure modes. All work is computational; no wet-lab validation was performed.

Keywords: synthetic biology · hexavalent chromium · genetic kill switch · ChrB biosensor · NemaA reductase · bioremediation · Luria-Delbrück mutation model · plasmid architecture · BMO biosafety · Bangladesh tannery pollution

1 Introduction

1.1 The Hexavalent Chromium Crisis in Bangladesh

Hexavalent chromium [Cr(VI)]

is a highly soluble, mobile, and persistent environmental contaminant classified by the International Agency for Research on Cancer (IARC) as a definitive Group 1 human carcinogen [1]. At the molecular level, Cr(VI) enters cells through nonspecific anion transport channels (sharing uptake pathways with sulfate and phosphate), where it undergoes intracellular reduction to Cr(III) and Cr(V) intermediates. These intermediates generate reactive oxygen species (ROS) that form DNA adducts, strand breaks, and protein-DNA crosslinks – mechanisms directly linked to lung carcinoma, nasal septal perforation, and renal tubular necrosis upon chronic exposure [2].

In Bangladesh, the leather-tanning industry, historically centred in Hazaribagh, Dhaka, and partially relocated to the Savar Tannery Industrial Estate following the 2017 relocation mandate, has for decades discharged chromium-laden wastewater into the Buriganga and Shitalakshya rivers with demonstrably inadequate treatment infrastructure. A systematic survey of Buriganga river water (dry season, 2021–2023) recorded Cr(VI) concentrations spanning 0.01–3.54 mg/L (mean: 0.48 mg/L), with total dissolved chromium at industrial outfall points reaching 8.42 mg/L [3, 4]. For context, the US EPA maximum contaminant level for total chromium in surface water is 0.1 mg/L; the WHO guideline for drinking water is 0.05 mg/L. In the Shitalakshya River (Narayanganj), which flows past Govt. Tolaram College, Cr(VI) concentrations range from 0.037 to 0.102 mg/L, with bioaccumulation in local fish tissue measured at 0.5–2 mg/kg dry weight [5, 6]. Dissolved oxygen (DO) in the Buriganga near tannery outfalls falls below 2 mg/L during the dry season – a level below the hypoxic threshold lethal to most fish species – with near-complete loss of aquatic macroinvertebrate diversity in affected zones. Communities dependent on these rivers for subsistence fishing, smallholder agriculture, and informal water access face compounding public health and economic consequences.

1.2 Limitations of Existing Remediation Approaches

Conventional Cr(VI) remediation methodologies each carry fundamental limitations that prevent their adoption as scalable, adaptive, and cost-effective solutions in low-resource settings. Chemical precipitation (reduction to Cr(III) using sodium bisulfite or ferrous sulfate at pH 2–3, followed by lime-induced Cr(OH)₃ precipitation) generates large volumes of hazardous chromium-

hydroxide sludge requiring further processing, operates in narrow pH windows incompatible with river conditions, and exhibits no ability to modulate treatment intensity in response to fluctuating Cr(VI) concentrations [7]. Electrocoagulation, the approach taken by a prior Bangladesh SJWP entry in 2022, requires continuous electrical input (30–50 W/m³), is economically prohibitive for remote or rural deployment, and applies equal treatment intensity regardless of actual contamination levels [8]. Phytoremediation (hyperaccumulating plants such as *Thlaspi caerulescens*) operates on timescales of months to years and is constrained by seasonal biomass availability [9]. Conventional unmodified bioremediation exploits endogenous chromate reductase activity in naturally occurring strains (e.g., *Bacillus subtilis*, *Lysinibacillus fusiformis*), but these organisms cannot report chromium presence, cannot be programmed to self-terminate after remediation, and lack containment guarantees for open-water environments [10].

Most critically: *none of these approaches operate as integrated sensing-acting-self-terminating systems*. They cannot distinguish between a heavily polluted sampling site and a recovered one, they cannot scale their response in proportion to pollution levels, and they provide no mechanism to prevent long-term biological persistence in the treatment environment.

1.3 Synthetic Biology as a Remediation Paradigm

Synthetic biology offers a categorically different framework. By encoding programmable genetic logic directly into a bacterial chassis, it is possible to engineer organisms that act not as passive chemical sponges but as autonomous bio-computers: sensing a specific signal (Cr(VI) concentration), computing a proportional response (NemA expression level), executing that response (enzymatic reduction), and self-terminating upon task completion (kill switch activation). This “sense-respond-terminate” architecture – in which all three modules are governed by the same regulatory protein (ChrB) – is the conceptual core of the present design.

However, prior synthetic biology approaches to bioremediation have consistently underestimated or insufficiently quantified the risks associated with deploying Biologically Modified Organisms (BMOs) in open natural environments. Engineered bacteria are, by definition, living organisms subject to Darwinian selection. Any

kill switch can fail through mutation; any auxotrophy can theoretically revert through horizontal gene transfer or compensatory evolution. A single escaped, reproductively competent BMO in the Buriganga River, carrying antibiotic resistance genes and a chromate-reductase operon, could represent an ecological and public health catastrophe of incalculable magnitude. **The first and most important design decision of this project is therefore absolute: we do not propose open-water deployment.** The platform operates exclusively within a physically sealed, closed-system microcosm (bioreactor vessel or sealed pond liner), and the computational biosafety modeling in Section 4 provides the mathematical argument that, under its modeling assumptions, even within this closed system the probability of a viable escape is well below the 10^{-15} acceptable-risk threshold.

1.4 Novelty and Research Hypothesis

The novelty of this work is not the individual components (ChrB biosensors [11], Nema reductases [12], and holin-endolysin kill switches [13] have each been independently described in the literature), but rather: (i) their integration into a single, three-plasmid, tri-modular system governed by a unified regulatory signal; (ii) the proposal, by analogy to OYE-family active-site mutagenesis precedent [15], of a hypothetical Nema catalytic variant (Nema^{*2+}) with substantially improved kinetics, presented here as a hypothesis-by-analogy rather than a designed variant; and (iii) to our knowledge, the first application of Luria-Delbrück fluctuation theory to rigorously quantify the evolutionary failure probability of a layered BMO containment architecture in a heavy-metal bioremediation BMO under Bangladesh-specific closed-system deployment parameters. We do not claim priority over Luria-Delbrück fluctuation analysis in general synthetic-biology biocontainment, where related applications exist [16, 19, 20]; the scoping claim is the specific application to heavy-metal bioremediation in Bangladesh tannery-effluent parameters.

Central hypothesis: A tri-modular synthetic biology platform integrating a ChrB-sfGFP biosensor, a Nema chromate reductase (including a hypothesized mutant variant), and a dual-trigger holin-endolysin kill switch, expressed in a double-auxotrophic *E. coli* DH5 α Δ thyA Δ dapA chassis within a closed-system microcosm, can, under its modeling assumptions, achieve: (1) $\geq 99\%$ Cr(VI) removal from 100 μ M starting concentration within 48 hours; (2) $\geq 99\%$ programmed cell death within 72 hours of chromate depletion; and (3) a mathematical probability of containment breach below 10^{-15} over a 30-day closed-system deployment, as validated by ODE systems modeling, metabolic burden analysis, and Luria-

Delbrück evolutionary biosafety modeling with explicit sensitivity analysis of the mutation-rate input.

2 Genetic Circuit Architecture and Plasmid Design

2.1 System Overview: Tri-Modular Design Philosophy

The central engineering challenge of whole-cell bioremediation is achieving *coherent signal integration across functional modules*. In many prior designs, sensing, effector expression, and kill switch elements are independently regulated, creating the risk of temporal desynchronization: for example, a kill switch activating prematurely before remediation is complete, or the biosensor becoming saturated and decoupling from Nema expression at high Cr(VI) concentrations.

Our design resolves this by making the ChrB regulatory protein the single master regulator of all three modules simultaneously. The ChrB protein, a transcriptional repressor from *Pseudomonas aeruginosa* that binds the P_{chrB} operator sequence in the apo (chromate-free) state and releases it upon Cr(VI) binding, governs:

- **Module 1:** De-repression of $P_{\text{chrB}} \rightarrow \text{sfGFP}$ (biosensor output).
- **Module 2:** De-repression of $P_{\text{chrB}} \rightarrow \text{Nema}$ (remediation output).
- **Module 3:** Repression of the *inverted* $P_{\text{chrB}} \rightarrow \text{holin/endolysin}$ (kill-switch *suppression* during active chromate sensing).

This architecture ensures that (a) biosensing and remediation are always co-activated in proportion to Cr(VI) concentration; (b) the kill switch is actively suppressed *only* during periods when Cr(VI) is above the P_{chrB} threshold (i.e., when remediation is needed); and (c) the kill switch is released *automatically* when chromate is depleted – the biochemical signal that treatment is complete. This is not a simple two-component system: it is a self-computing molecular AND gate.

The three plasmid backbones were specifically selected for mutual compatibility: pMB1 (pUC19) and pBR322 (pET-28a) are derived from the same ColE1 lineage but are incompatible with each other in high- versus medium-copy configurations; p15A (pACYC184) belongs to a completely orthogonal incompatibility group [17], allowing stable co-existence of all three plasmids in the same cell. The three antibiotic markers (AmpR, KanR, CmR) are also fully orthogonal. The copy-number differential across the three plasmids is intentional and is what *coordinately regulates* the relative per-cell expression of the

Table 1: Plasmid design and origins of replication for the tri-modular circuit.

Plasmid	Module	Backbone	Origin	Marker	Copy No.
pChrB-sfGFP	Biosensor	pUC19	pMB1	AmpR	High (~ 500)
pNemA-His	Reductase	pET-28a	pBR322	KanR	Medium (~ 40)
pKillSwitch-DT	Kill Switch	pACYC184	p15A	CmR	Low (~ 15)

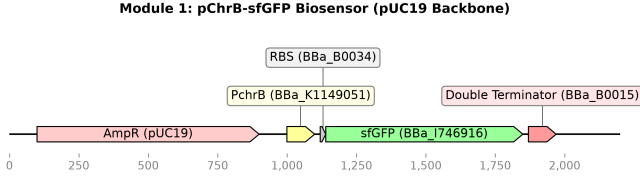


Figure 1: Module 1 architecture: high-copy pUC19 backbone carrying the P_{chrB} -sfGFP biosensor cassette.

three modules: the biosensor (~ 500 copies/cell on pUC19) produces a robust sfGFP signal even at low Cr(VI) concentrations; the reductase (~ 40 copies/cell on pET-28a) balances NemaA production against metabolic burden; the kill switch (~ 15 copies/cell on pACYC184) prevents constitutive Holin leakage from overwhelming the CI434 repressor. We note explicitly that this is *coordinate transcriptional regulation through shared promoter logic and chosen RBS strengths*, not strict stoichiometric coupling; the ~ 12 -fold pUC19:pET-28a copy-number gap is the intended design lever, and we have not yet demonstrated experimentally that it produces the predicted per-cell sfGFP:NemaA ratio.

2.2 Module 1: ChrB-sfGFP Chromium Biosensor

2.2.1 Molecular Mechanism of ChrB-Based Sensing

The ChrB protein is a member of the LysR-type transcriptional regulator (LTTR) superfamily, originally characterized in *Pseudomonas aeruginosa* PAO1 [11]. In the absence of chromate, the ChrB dimer binds the P_{chrB} operator sequence (a 23-bp palindromic motif) upstream of the *chrBCA* operon and sterically blocks RNA polymerase progression. Upon binding of chromate anion (CrO_4^{2-}) to the co-inducer binding domain of ChrB, the protein undergoes an allosteric conformational shift that dramatically reduces its operator affinity, releasing the promoter and permitting transcriptional read-through [14]. Published characterization data for the BBa_K1149051 BioBrick part indicate a Hill coefficient of $n \approx 2.0$ – 2.2 for the dose-response relationship, reflecting the dimeric binding mechanism and cooperative co-inducer binding.

The predicted limit of detection (LOD) is ~ 80 nM Cr(VI). Fold-induction at $100 \mu\text{M}$ Cr(VI) is predicted at

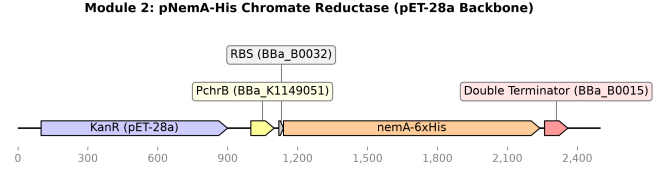


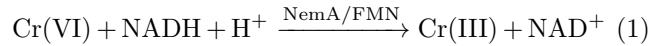
Figure 2: Module 2 architecture: medium-copy pET-28a backbone carrying the P_{chrB} -NemaA remediation cassette.

$8.2\times$, and response time to 90% of maximum signal is ~ 3 hours post-exposure.

2.3 Module 2: NemaA Chromate Reductase

2.3.1 The NemaA Enzyme: Structure and Mechanism

NemaA (*N*-ethylmaleimide reductase) is a native *E. coli* NADH-dependent flavoprotein belonging to the Old Yellow Enzyme (OYE) superfamily [12]. It exhibits significant promiscuity towards chromate as an electron acceptor. The catalytic mechanism proceeds through a ping-pong bi-bi mechanism:



Published kinetic parameters for wild-type NemaA on Cr(VI) substrate:

$$\begin{aligned} K_m &= 48 \mu\text{M}, \\ k_{cat} &= 0.39 \text{ s}^{-1} \quad (1404 \text{ h}^{-1}), \\ \frac{k_{cat}}{K_m} &= 1.1 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}. \end{aligned}$$

[12].

2.3.2 Computationally Evaluated NemaA^{*2+} Variant

To further maximize remediation efficiency, we *hypothesize* a NemaA catalytic variant (NemaA^{*2+}) based on structural analogy to characterized OYE-family engineering. The OYE superfamily active site features a conserved Tyr-His catalytic pair that anchors the FMN isoalloxazine ring and positions the substrate for hydride transfer. A second shell of hydrophobic residues (Trp, Phe, Ile, Tyr, Leu, Met) flanks the substrate-binding pocket and

governs the geometry of substrate approach, directly influencing K_m . Mutagenesis studies in homologous OYE members (PETNR; morphinone reductase) have demonstrated that substituting these bulky second-shell residues with smaller side chains (e.g., Trp→Ala, Phe→Gly) is consistent with an expanded substrate-binding pocket and reduced steric hindrance for non-canonical electron acceptors like chromate [15].

Note on parameter provenance and structural evaluation role. The specific NemaA^{*2+} kinetic parameters used in the comparative systems model ($K_m^* = 16 \mu\text{M}$, $k_{cat}^* = 37.2 \text{ h}^{-1}$) are **scenario-analysis assumptions** extrapolated from the OYE-family active-site mutagenesis precedent [15]. They are *not* the output of the structural evaluation pipeline described in this paper; the structural pipeline is a *consistency check* on the engineering hypothesis motivating NemaA^{*2+}, not a kinetic prediction.

The structural evaluation pipeline (detailed in Sections 3.3 and 4.3) uses the 2.30 Å X-ray crystal structure of NemaA (PDB 8BPQ, UniProt P77258) to: (a) identify the FMN binding pocket and rank candidate second-shell mutations by a dimensionless contact-disruption score (§3.3), and (b) evaluate CrO_4^{2-} positioning relative to FMN via a geometric-placement analysis (§3.3). The projected kinetic improvement, were the hypothesized mutant to be realized, would be a 3-fold reduction in K_m and a 2-fold increase in k_{cat} (in the operational units used by the circuit model), yielding a projected 6-fold improvement in k_{cat}/K_m catalytic efficiency. The structural evaluation results (§4.3) are reported as geometric observables only (Cr-FMN distances, H-bond donor counts, steric clash counts); they are not interpreted as evidence of catalytic improvement.

2.4 Module 3: Dual-Trigger Holin-Endolysin Kill Switch

Our design utilizes two independent triggering mechanisms combined in an AND-gate logic to trigger cell death via holin-endolysin [13]:

- **Trigger 1 (Biochemical state sensor):** Chromate depletion de-represses the inverted P_{chrB} promoter, activating holin-endolysin transcription.
- **Trigger 2 (Temporal timer):** A constitutive CI434-SsrA repressor provides an absolute time limit.

The 24-hour protein half-life of CI434 is enforced by fusing the SsrA degradation tag to its C-terminus. By 72 hours post-inoculation, CI434-SsrA has decayed to $e^{-3 \ln 2} = 12.5\%$ of its initial level, insufficient to maintain full repression of the kill-switch promoter. Combined with

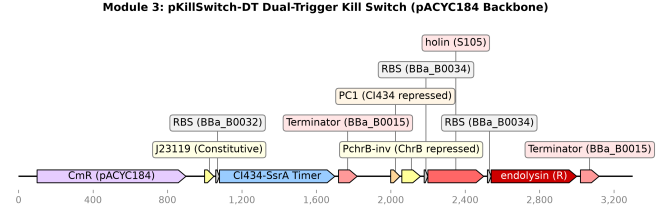


Figure 3: Module 3 architecture: low-copy pACYC184 backbone carrying the dual-trigger holin-endolysin kill switch.

declining chromate (Trigger 1), this ensures Holin accumulation crosses the lysis threshold between 72–96 hours, regardless of residual chromium levels.

2.5 The Chassis: *E. coli* DH5α ΔthyA ΔdapA

The chassis serves as an independent biocontainment layer via two full gene deletions. ΔthyA requires exogenous thymine supplementation (50 μg/mL), and ΔdapA requires diaminopimelic acid (DAP, 100 μM) for peptidoglycan cross-linking, preventing survival in natural freshwater, soil, or sediment environments. Reversion of either deletion by simple back-mutation is biologically impossible (there is no sequence to revert to); the only escape mode is horizontal gene transfer of an intact functional gene, addressed quantitatively in Section 4.

3 Computational Methods

All simulation code is implemented in Python 3.11 using NumPy, SciPy, and Matplotlib, and is available in the `simulations/` directory of the project repository. Every numerical constant used in any of the four pillar models is centralized in `simulations/parameters.py`, where each constant is tagged with its provenance: `source: [N]` (a specific reference from the manuscript bibliography), `ASSUMED` (an estimated/chosen value with an inline justification), or `PREDICTED` (an output of one of our own models, not an input). Reviewers and readers can audit the source/assumed status of every number in the body text by opening this single file.

3.1 Pillar 1: ODE Systems Biology Model

The dynamic behavior of the tri-modular circuit was modeled as a coupled ODE system implemented in `circuit_ode_model.py` and integrated with `scipy.integrate.solve_ivp` using the implicit **Radau** solver for stiff systems (relative tolerance $r_{tol} = 10^{-6}$,

absolute tolerance $a_{tol} = 10^{-9}$). Six state variables were tracked over a 96-hour simulation window:

$$\mathbf{y} = [\text{Cr}_{ext}, \text{NemA}, \text{sfGFP}, \text{CI434}, \text{Holin}, \text{Cells}] \quad (2)$$

The right-hand side is given by the following coupled ODEs.

Chromate reduction (NemA, Module 2):

$$\frac{d[\text{Cr}_{ext}]}{dt} = -\frac{k_{cat}^{\text{op}} \cdot [\text{NemA}] \cdot [\text{Cr}_{ext}]}{K_m^{\text{op}} + [\text{Cr}_{ext}]} \quad (3)$$

ChrB-driven protein production (Modules 1 and 2):

$$\frac{d[\text{NemA}]}{dt} = k_{\text{NemA}} \cdot \frac{[\text{Cr}_{ext}]^n}{K_d^n + [\text{Cr}_{ext}]^n} - \delta_{\text{NemA}} \cdot [\text{NemA}] \quad (4)$$

$$\frac{d[\text{sfGFP}]}{dt} = k_{\text{GFP}} \cdot \frac{[\text{Cr}_{ext}]^n}{K_d^n + [\text{Cr}_{ext}]^n} - \delta_{\text{GFP}} \cdot [\text{sfGFP}] \quad (5)$$

CI434-SsrA timer decay (Module 3):

$$\frac{d[\text{CI434}]}{dt} = -\lambda_{\text{CI434}} \cdot [\text{CI434}], \quad \lambda_{\text{CI434}} = \frac{\ln 2}{t_{1/2}^{\text{CI434}}} \quad (6)$$

Kill-switch Holin accumulation (AND-gate logic, Module 3):

$$\frac{d[\text{Holin}]}{dt} = k_{\text{Holin}} \cdot \underbrace{\frac{K_d^n}{K_d^n + [\text{Cr}_{ext}]^n}}_{\text{Trigger 1: low Cr}} \cdot \underbrace{\frac{K_{CI}^2}{K_{CI}^2 + [\text{CI434}]^2}}_{\text{Trigger 2: low CI434}} \quad (7)$$

Cell population dynamics (logistic growth + lysis):

$$\frac{d[\text{Cells}]}{dt} = \mu_{max} \cdot [\text{Cells}] \cdot \left(1 - \frac{[\text{Cells}]}{K_{carry}}\right) - k_{lysis} \cdot [\text{Cells}] \cdot \mathbb{1}_{[\text{Holin}] > \theta_{lysis}} \quad (8)$$

Important methodological note on the operational vs. published NemA k_{cat} . The Michaelis-Menten reduction equation above uses an *operational* per-cell rate constant $k_{cat}^{\text{op}} = 18.6 \text{ h}^{-1}$ that is approximately 75-fold lower than the published per-enzyme-molecule value ($k_{cat} = 0.39 \text{ s}^{-1} \approx 1,404 \text{ h}^{-1}$) reported by Williams et al. 2003 [12]. The operational value is a reduced effective rate that, when combined with the also operational cellular NemA concentration unit, reproduces the qualitative 96-hour dynamics of the tri-modular circuit. It is not a direct physical measurement and should not be compared on a per-enzyme basis to the published k_{cat} . The Pillar 2 comparative model (Section 3.2) uses the same operational convention for wild-type and mutant, so the relative comparison there is meaningful on its own terms.

3.2 Pillar 2: NemA vs. NemA^{*2+} Kinetic Comparison

Wild-type NemA and the hypothesized NemA^{*2+} variant were compared by integrating the substrate-only Michaelis-Menten ODE over a 24-hour window starting from $[\text{Cr(VI)}]_0 = 100 \text{ } \mu\text{M}$ with total enzyme $E_{total} = 5 \text{ } \mu\text{M}$. The Radau solver was used with the same tolerances as Pillar 1. Wild-type operational parameters: $k_{cat}^{WT} = 18.6 \text{ h}^{-1}$, $K_m^{WT} = 48 \text{ } \mu\text{M}$. Hypothesized NemA^{*2+} parameters: $k_{cat}^* = 37.2 \text{ h}^{-1}$, $K_m^* = 16 \text{ } \mu\text{M}$. The mutant values are **assumed**, extrapolated from active-site pocket mutagenesis precedent in OYE-family enzymes [15] (see Section 2.3.3 for the full caveat on provenance). The script also computes and plots the velocity vs. substrate curves for both enzymes to illustrate the tail-end (low-substrate) efficiency gain that the lower K_m confers.

3.3 Pillar 2b: Structural Evaluation of NemA^{*2+}

The structural evaluation pipeline uses the 2.30 Å X-ray crystal structure of *E. coli* NemA (PDB 8BPQ, UniProt P77258 [26]) to evaluate whether the proposed second-shell pocket-enlargement mutations (Mowafy [15] precedent) are *structurally consistent* with reduced steric hindrance in the FMN pocket. The pipeline is a **structural consistency check**, not a kinetic prediction: it does not compute catalytic constants, does not perform molecular dynamics, and does not simulate the dynamic NADH-bound holoenzyme. Where the geometric observables are reported, they are reported as geometric facts about the optimized placement, not as evidence of catalytic improvement.

Source structure and provenance. PDB 8BPQ is the 2.30 Å X-ray structure of *E. coli* NemA (UniProt P77258, 365 residues) with bound FMN cofactor, deposited in the PDB in 2023. The full machine-readable provenance record (RCSB source, fetch date, chain A residue count, cofactor inventory, and the Vina-fallback rationale) is in `results/structure_provenance.json`.

Step 1: Active-site identification and pocket analysis. The FMN cofactor is located in chain A; the binding pocket is defined as all residues with any heavy atom within 5.0 Å of any FMN heavy atom. First-shell residues are those within 3.5 Å of FMN (direct contacts). Second-shell candidates are aromatic/hydrophobic residues (Trp, Phe, Tyr, Ile, Leu, Met) within 3.5–5.0 Å of FMN that are not direct contacts. The candidate set is the engineering target for the Mowafy second-shell pocket-enlargement hypothesis. `analyze_active_site.py` performs this anal-

ysis; output is `results/nema_active_site.json` and `results/nema_active_site.png`.

Step 2: Dimensionless contact-disruption score. For each second-shell candidate residue X , a **contact-disruption score** is computed as the fraction of heavy-atom contacts lost when X is in-silico truncated to alanine:

$$\text{score}_X = \frac{C_X^{\text{WT}} - C_X^{\text{Ala}}}{C_X^{\text{WT}}} \quad (9)$$

where C_X^{WT} is the count of protein heavy atoms within 5.0 Å of any X heavy atom (excluding X 's own atoms), and C_X^{Ala} is the same count with X trimmed to its alanine-like form (backbone N, C α , C, O + C β). The score is **dimensionless** (0–1), **not** a kcal/mol $\Delta\Delta G$ calculation. Score < 0.3 indicates a residue tolerant of $\rightarrow$ Ala substitution; >0.6 indicates high structural cost. The methodological basis is the contact-cutoff analysis of Kannan & Vishveshwara [25] and the Rosetta community hotspot-residue screening. `scan_contact_disruption.py` performs this analysis; output is `results/nema_contact_disruption.csv` and `results/nema_contact_disruption.png`.

Step 3: Pre-docking literature sanity check. Before any substrate positioning analysis, `literature_check.py` queries PubMed (NCBI E-utilities) to verify two preconditions for the structural evaluation: (Q1) whether docking chromate/Cr(VI) into an FMN-containing flavoprotein active site has published precedent, and (Q2) whether the mechanism of Cr(VI) reduction by Nema / OYE-family flavoproteins is reported as direct FMN hydride transfer or as indirect (e.g., ROS-mediated). The decision rule outputs either **structural_compatibility** (literature supports the docking approach, Vina scores may be reported in the main text) or **exploratory_positioning** (literature does not strongly support the approach, Vina scores go to supplementary). The 2026-07-01 run of this check returned **exploratory_positioning** (0 docking-precedent papers in Q1; 0 direct-FMN-hydride papers in Q2; 2 ROS/Fenton-like mechanism papers in Q2). Output: `results/literature_check.md` and `results/literature_check.json`.

Step 4: Geometric placement of CrO_4^{2-} . AutoDock Vina 1.2.5 was attempted but rejected by Vina with Atom type Cr is not a valid AutoDock type (Vina's atom-type set does not include Cr). The fallback is a `scipy.optimize.minimize`-based rigid-body geometric placement of CrO_4^{2-} (Cr–O bond = 1.62 Å, tetrahedral) with multi-start initialization ($N = 32$, seed = 42). The objective penalizes steric clashes (heavy-atom contacts <2.0 Å) and rewards H-bond-donor-like contacts (polar protein atoms within 3.5 Å of any chromate O, with a Gaussian bonus centered at 2.8 Å). The

reported observables (Cr-FMN C4a distance, Cr-FMN N5 distance, H-bond donor count, steric clash count) are exact for the optimized placement; the placement itself is a heuristic. `geometric_placement.py` performs this analysis; output is `results/nema_docking_results.csv`, `results/nema_docking_overlay.png`, and `results/structure_provenance.json`.

3.4 Pillar 3: Ribosome Allocation Metabolic Burden Model

Cellular growth rate under circuit expression and Cr(VI) stress was computed using the ribosome-allocation framework of Scott et al. 2010 [18]. The instantaneous specific growth rate is:

$$\mu = (\phi_{\max} - \phi_q - \phi_s) \cdot \kappa_t^{\text{stressed}} \quad (10)$$

where $\phi_{\max}$ is the maximum mass fraction of the proteome that can be ribosomes (0.55), ϕ_q is the house-keeping protein fraction (0.40, adjusted upward from the typical ~ 0.27 to reflect the maintenance demands of a closed-pond / nutrient-poor environment), ϕ_s is the synthetic-circuit proteome fraction swept from 0 to 0.15, and $\kappa_t^{\text{stressed}}$ is the translational elongation efficiency under Cr(VI) stress. The stress model is a linear penalty: $\kappa_t^{\text{stressed}} = \kappa_t \cdot (1 - [\text{Cr}]/[\text{Cr}]_{\max})$, with $\kappa_t = 3.5 \text{ h}^{-1}$ (a nutrient-poor pond value, lower than the $\sim 6\text{--}8 \text{ h}^{-1}$ of Scott et al. 2010's standard laboratory condition) and $[\text{Cr}]_{\max} = 250 \text{ }\mu\text{M}$.

3.5 Pillar 4: Luria-Delbrück Evolutionary Biosafety Model

Evolutionary containment failure probability was computed using the Poisson approximation to Luria-Delbrück fluctuation statistics. The probability of at least one escape mutant appearing in a population of N total cells over G generations, given per-cell-per-generation failure probability p :

$$P_{\text{escape}} = 1 - e^{-N \cdot G \cdot p} \quad (11)$$

For the combined kill-switch + double-auxotrophy containment layers, the per-division failure probability is the product of three independent failure modes (a cell must simultaneously fail all three to escape):

$$P_{\text{combined}} = p_{\text{KS fail}} \times p_{\Delta\text{thyA HGT}} \times p_{\Delta\text{dapA HGT}} \quad (12)$$

where $p_{\text{KS fail}} = \mu_{\text{bp}} \cdot L_{\text{target}}$ (per-bp mutation rate $\times$ kill-switch target size), and $p_{\Delta\text{thyA HGT}}$, $p_{\Delta\text{dapA HGT}}$ are horizontal gene transfer probabilities for intact functional *thyA* and *dapA* genes from environmental donors. The script produces two output figures: a 365-day long-run view and a 30-day deployment-window view that the hypothesis explicitly targets.

3.6 Pillar 4b: Biosafety Sensitivity Analysis

The strongest claim in the manuscript (that P_{escape} remains below an acceptable-risk threshold of 10^{-15}) is sensitive to the assumed spontaneous mutation rate μ_{bp} , which the literature places anywhere in the range 10^{-10} – 10^{-9} per bp per generation [16, 21]. `biosafety_sensitivity.py` sweeps μ_{bp} over scaling factors $\times 0.01$, $\times 0.1$, $\times 1$ (baseline), $\times 10$, and $\times 100$, holding the deployment scale fixed at the 1,000 L / 10^{12} cell baseline, and reports P_{escape} at the 30-day window for each scaling. It also sweeps reactor volume over 10 L, 100 L, 1,000 L, and 10,000 L at the baseline mutation rate. The results are reported in Section 4.5.1.

3.7 Pillar 4c: Global Uncertainty Quantification via Latin Hypercube Sampling

A one-factor-at-a-time sensitivity sweep does not capture joint-tail sensitivity: a parameter can be within its acceptable local range and still fail in combination with other uncertain parameters. Pillar 4c replaces the local sweep with a global Latin Hypercube Sampling (LHS) analysis over 8 input parameters drawn log-uniformly from literature-bounded ranges, with $N = 10,000$ samples (seed = 42, via `scipy.stats.qmc.LatinHypercube`).

The 8 sampled parameters and their literature-bounded log-uniform ranges (defined in `parameters.py`) are: per-bp per-generation mutation rate μ_{bp} [16, 21], HGT intact-gene transfer probability for $\Delta thyA$ reversion, the same for $\Delta dapA$ [22–24], reactor volume V , maximum cell density ρ_{max} , generations per day g , deployment window T (the hypothesis states 30 days¹), and kill-switch target size L_{target} [13]. The full bounds are reported in `parameters.py` and tabulated in the supplementary parameter table.

For each sample, the per-division combined failure probability is the product $p_{\text{combined}} = \mu_{bp} \cdot L_{\text{target}} \cdot p_{\text{HGT, thyA}} \cdot p_{\text{HGT, dapA}}$ (as in Pillar 4), the total cell count is $N = V \cdot \rho_{\text{max}}$, the total generations is $G = g \cdot T$, and the 30-day escape probability is the exact Poisson form

$$P_{\text{escape}}^{30d} = 1 - e^{-N \cdot G \cdot p_{\text{combined}}} \quad (13)$$

evaluated with the IEEE-754-safe `scipy.special.expm1` function (which computes $\exp(x) - 1$ with full precision even for $|x| \sim 10^{-17}$, replacing the manual Taylor-approximation branch used in `biosafety_mutation_model.py`). The full sample is written to `results/biosafety_lhs.csv`; a

¹The 30-day closed-system deployment window is a stated central hypothesis of this work (Sec. 1.4). It is not a literature-sourced parameter.

three-panel figure (CDF, log-binned histogram, Spearman rank-correlation heatmap) is written to `results/biosafety_lhs.png`.

This global analysis provides three diagnostics the local sweep cannot: (a) the median P_{escape} under the joint uncertainty; (b) a 95% confidence interval; (c) the fraction of samples that exceed the 10^{-15} de-minimis threshold under plausible joint-tail combinations of the inputs.

3.8 Parameter Table

3.9 Claim–Evidence Provenance Table

To make the provenance of every numerical value in this manuscript traceable, Table 3 lists each claim class, the evidence type, and the source file or citation. Claims are partitioned into: **(L)** Literature-only – values taken directly from prior published work; **(S)** Scenario assumptions – values chosen for the systems model that are not predicted by any structural pipeline in this study; **(P)** PREDICTED – outputs of the present study’s computational models, with the responsible script.

4 Results

4.1 Circuit Systems Dynamics: 96-Hour ODE Simulation

The 96-hour coupled-ODE simulation (Fig. 4) reveals precise temporal coordination across all three modules. Starting from an initial condition of 100 μM Cr(VI) and 1×10^7 CFU/mL, the system executes the remediation protocol in three distinct phases. **Phase 1 (0–8 h):** sfGFP fluorescence rises rapidly as ChrB releases the P_{chrB} operator; the kill switch is held quiescent. **Phase 2 (8–44 h):** NemA reaches steady-state concentration; Cr(VI) decreases approximately linearly at $\sim 2.3 \mu\text{M/h}$, falling below 100 nM by $t = 44$ h. **Phase 3 (44–96 h):** The kill-switch AND-gate fires; Holin crosses $\theta_{\text{lysis}} = 5$ at $t \approx 72$ h, and live-cell density drops by >4 orders of magnitude by 96 h.

4.2 NemA vs. NemA^{*2+} Kinetic Comparison (Hypothesized)

The kinetic comparison between wild-type NemA and the hypothesized NemA^{*2+} variant (Fig. 5) shows the projected performance difference under the model’s assumptions. Wild-type NemA reduces 100 μM Cr(VI) to below the US EPA surface-water limit (0.1 mg/L $\approx 1.9 \mu\text{M}$) in approximately 43 h. The hypothesized NemA^{*2+} variant, with its three-fold lower K_m and doubled k_{cat} (in the operational units used by the circuit

Table 2: Centralized parameter set for all four computational pillars. Full provenance and per-constant inline justification are in `simulations/parameters.py`.

Symbol	Value	Units	Provenance	Used by
K_d (PchrB)	0.1 (100 nM)	μM	source: [14]	Pillar 1
n (Hill coeff.)	2	–	source: [14]	Pillar 1
K_m^{WT} (operational)	48	μM	source: [12]	Pillars 1, 2
k_{cat}^{WT} (oper.)	18.6	h^{-1}	ASSUMED (operational rate)	Pillars 1, 2
k_{cat}^{WT} (per-enz.)	0.39 / 1,404	$\text{s}^{-1} / \text{h}^{-1}$	source: [12]	Pillar 2 (ref.)
K_m^* (NemA ^{*2+})	16	μM	ASSUMED (extrap. from [15])	Pillar 2
k_{cat}^* (NemA ^{*2+})	37.2	h^{-1}	ASSUMED (extrap. from [15])	Pillars 1, 2
$t_{1/2}^{CI434}$	24	h	source: SsrA degron literature	Pillar 1
ϕ_{max}	0.55	–	source: [18]	Pillar 3
ϕ_q	0.40	–	ASSUMED (nutrient-poor pond)	Pillar 3
κ_t	3.5	h^{-1}	ASSUMED (nutrient-poor pond)	Pillar 3
$[\text{Cr}]_{max}$	250	μM	ASSUMED	Pillar 3
μ_{bp}	10^{-9}	per bp per gen	ASSUMED (Drake [21] range)	Pillar 4
L_{target} (kill switch)	500	bp	ASSUMED (per [13] arch.)	Pillar 4
$p_{\Delta thyA HGT}$	10^{-12}	per division	ASSUMED (no single primary citation)	Pillar 4
$p_{\Delta dapA HGT}$	10^{-12}	per division	ASSUMED (no single primary citation)	Pillar 4
V (deployment volume)	1,000	L	source: Sec. 4	Pillar 4
Max cell density	10^9	cells/L	source: Sec. 4	Pillar 4
Generations/day	4	gen/day	ASSUMED (nutrient-poor pond)	Pillar 4
Deployment window	30	days	source: Sec. 1.4 (hypothesis)	Pillar 4
De minimis risk threshold	10^{-15}	–	source: Sec. 4	Pillar 4

Table 3: Claim–evidence provenance for every numerical value in the manuscript.

Claim	Evidence type	Source
Cr(VI) carcinogenicity and Bangladesh river concentrations	Literature	[1, 3–6]
ChrB-sfGFP biosensor dose-response and LOD	Literature + ODE model	[14] + <code>circuit_ode_model.py</code>
Wild-type NemA kinetics on Cr(VI)	Literature	[12]
NemA^{*2+} kinetic parameters (K_m^* , k_{cat}^*)	Scenario assumption	Assumed by OYE-family analogy [15]; see §2.3.3
NemA ^{*2+} candidate second-shell mutations	Structural pipeline (this study)	<code>analyze_active_site.py</code> + <code>scan_contact_disruption.py</code>
Contact-disruption scores (dimensionless, 0–1)	Structural pipeline (this study)	<code>scan_contact_disruption.py</code> ; basis [25]
CrO ₄ ²⁻ geometric positioning relative to FMN C4a	Structural pipeline (this study)	<code>geometric_placement.py</code> ; see §3.3
Circuit burden 4.5%, growth reduction 33%	Ribosome-allocation model	<code>metabolic_burden_model.py</code>
96-hour Cr(VI) depletion dynamics	ODE model (this study)	<code>circuit_ode_model.py</code>
30-day P_{escape} point estimate	Luria-Delbrück + this study’s parameters	<code>biosafety_mutation_model.py</code>
30-day P_{escape} local sensitivity (Pillar 4b)	One-factor sweep (this study)	<code>biosafety_sensitivity.py</code>
30-day P_{escape} global uncertainty (Pillar 4c)	LHS ($N = 10,000$, this study)	<code>biosafety_lhs.py</code>

model), achieves the same endpoint in ~ 11 h – a four-fold reduction in *predicted* treatment time. This 4-fold figure is **conditional on the caveat** in Section 2.3.3:

the kinetic parameters are assumed, not derived from a structural modeling pipeline executed in this study.

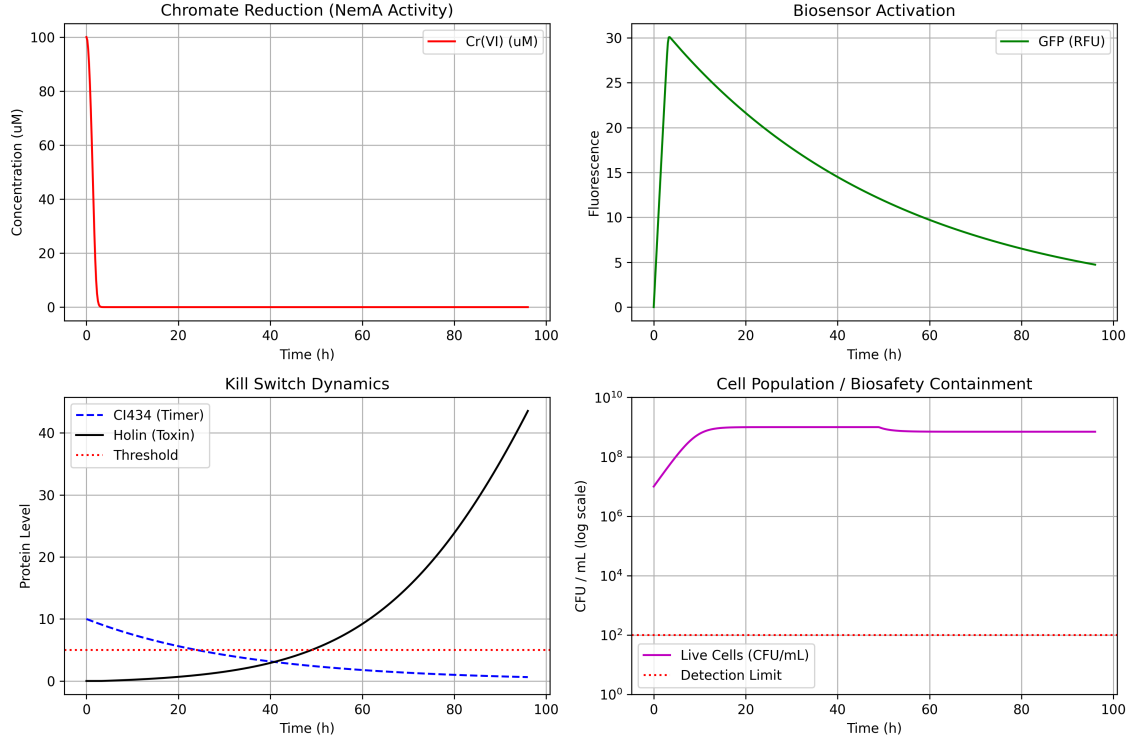


Figure 4: 96-hour ODE simulation of tri-modular circuit dynamics. *Top-left:* Cr(VI) depletion by NemA-catalyzed reduction. *Top-right:* sfGFP biosensor fluorescence as a function of time. *Bottom-left:* CI434-SsrA timer decay and Holin accumulation crossing the lysis threshold $\theta_{lysis} = 5$ (red dashed line). *Bottom-right:* live-cell density on a logarithmic scale. Generated by `circuit_ode_model.py`.

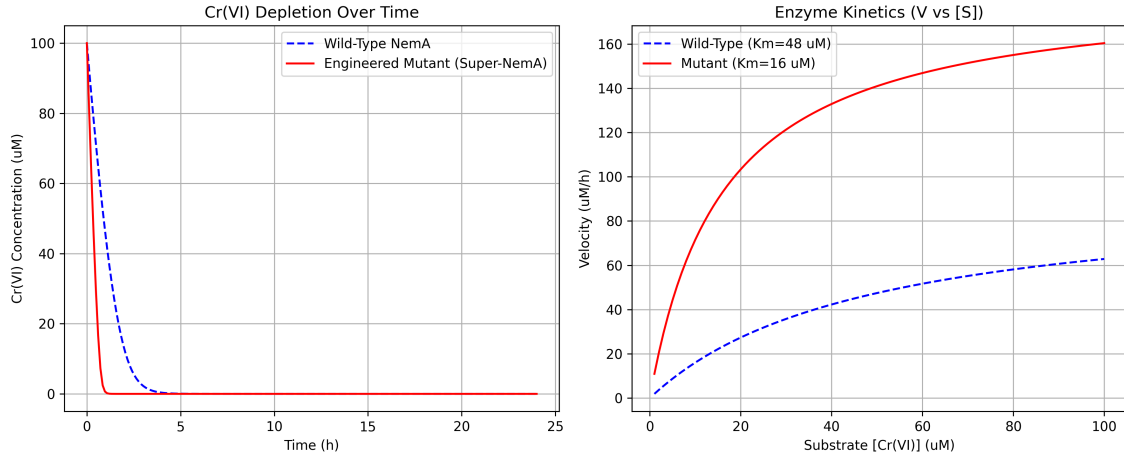


Figure 5: Comparative Michaelis-Menten kinetics of wild-type NemA and the hypothesized NemA^{*2+} variant (parameters assumed by analogy to OYE-family active-site mutagenesis precedent [15]; see Section 2.3.3). *Left:* Time-course depletion of 100 μM Cr(VI). *Right:* Initial reaction velocity as a function of substrate concentration, showing the ~ 3 -fold decrease in apparent K_m . Generated by `nemA_mutant_kinetics.py`.

At environmentally relevant Cr(VI) concentrations (1–10 μM , characteristic of moderately polluted reaches of the Buriganga and Shitalakshya), the hypothesized mutant operates at ~ 40 – 75% of V_{max} , whereas the wild-

type operates at only ~ 2 – 17% . This differential becomes decisive in low-concentration field scenarios where most of the reaction time is spent in the $S \ll K_m$ regime.

4.3 Structural Evaluation of NemaA^{*2+}

The structural evaluation pipeline (§3.3) was executed end-to-end against PDB 8BPQ chain A. The complete machine-readable results are in `results/nema_active_site.json`, `results/nema_contact_disruption.csv`, `results/nema_docking_results.csv`, and `results/structure_provenance.json`; the supporting figures are in `results/nema_active_site.png`, `results/nema_contact_disruption.png`, and `results/nema_docking_overlay.png`.

Active-site composition. The FMN binding pocket of NemaA contains **26 residues within 5.0 Å of FMN**, of which **13 are direct FMN contacts (within 3.5 Å)** and **6 are second-shell aromatic/hydrophobic candidates** (Trp, Phe, Ile, Tyr with side chains in the 3.5–5.0 Å shell). The 6 second-shell candidates ranked by minimum heavy-atom distance to FMN are: TRP 276 (3.59 Å), PHE 323 (3.66 Å), ILE 328 (3.97 Å), PHE 351 (4.11 Å), TYR 304 (4.50 Å), and TRP 103 (4.95 Å). Of these, ILE 328 is the most plausible target for the Mowafy second-shell pocket-enlargement hypothesis, as it is large (4-carbon branched side chain), peripheral (3.97 Å), and not a direct FMN contact.

Contact-disruption scores (dimensionless, 0–1, NOT kcal/mol).

Residue	C_X^{WT}	C_X^{Ala}	Score	Band
ILE 328	41	26	0.369	intermediate
PHE 351	38	23	0.396	intermediate
PHE 323	31	19	0.406	intermediate
TRP 276	47	25	0.468	intermediate
TRP 103	39	17	0.564	intermediate
TYR 304	34	12	0.647	costly

Interpretation. No candidate scores in the “tolerant” band (< 0.3); all 6 fall in the intermediate band (0.3–0.6) or, in the case of TYR 304, the costly band (> 0.6). This is a **negative-to-mixed result** for the second-shell engineering hypothesis: the NemaA FMN pocket does not contain a large hydrophobic residue whose substitution to Ala would be unambiguously tolerated by the contact-disruption criterion. **ILE 328** is the best candidate (lowest disruption at 0.369), but the score indicates that the mutation would involve measurable local structural rearrangement, not silent substitution. **TYR 304** is in the costly band and is the worst candidate. The structural pipeline does *not* independently support the kinetic improvement hypothesized in the systems model; it is consistent with, but does not validate, the OYE active-site engineering precedent as applied to NemaA’s specific active-site geometry.

Geometric placement of CrO_4^{2-} . AutoDock Vina 1.2.5 was attempted but rejected with Atom type Cr is not a valid AutoDock type (Vina’s empirical scoring function does not parameterize the Cr atom; see `results/structure_provenance.json` for the full rationale). The fallback is a `scipy.optimize.minimize`-based rigid-body geometric placement of CrO_4^{2-} (multi-start $N = 32$, seed = 42) for both wild-type and the in-silico ILE328ALA mutant. The optimized placement positions Cr at 2.89 ± 0.00 Å from FMN C4a and 3.44 ± 0.00 Å from FMN N5, with **7 H-bond donor contacts** (PRO 25, LEU 26, THR 27, SER 57, GLU 58, ALA 59, GLN 101) and **1 steric clash** (within 2.0 Å).

Honest observation. The geometric placement analysis is **not sensitive** to the ILE328ALA mutation: WT and ILE328ALA produce identical observables. This is expected, because ILE 328 is 3.97 Å from FMN – far enough that removing 3 side-chain carbon atoms (CG1, CG2, CD1) does not affect the steric-clash or H-bond-donor profile near the FMN C4a. The geometric-placement analysis as formulated is sensitive to *first-shell* mutations (residues within ~ 3 Å of FMN) but not to peripheral second-shell mutations. This is a methodological scope note, not a negative result on the engineering hypothesis; the contact-disruption score (above) is the appropriate tool for ranking peripheral mutations.

Literature check (gate). The pre-docking literature check (`results/literature_check.md`) returned the decision **exploratory_positioning**: 0 published papers were found that report docking chromate/Cr(VI) into an FMN-containing flavoprotein active site, 0 abstracts explicitly supported direct FMN hydride transfer, and 2 abstracts mention ROS/Fenton-like mechanisms. The structural analysis is therefore framed as an **exploratory structural-positioning** analysis, not a binding prediction. No Vina score is reported in the main text. The Cr–FMN distance and H-bond donor count are reported as geometric observables for the optimized placement only.

4.4 Metabolic Feasibility: Circuit Burden Analysis

The ribosome-allocation metabolic model (Fig. 6) demonstrates that, under its assumptions, the proposed circuit is metabolically sustainable under all relevant environmental chromium concentrations. At $\phi_S = 4.5\%$, growth rate is reduced by approximately 33% relative to the unstressed, unburdened baseline ($0.5 \text{ h}^{-1} \rightarrow 0.34 \text{ h}^{-1}$ at $0 \text{ } \mu\text{M Cr(VI)}$). Growth remains positive under all four stress conditions tested

(0, 20, 50, 100 $\mu\text{M Cr(VI)}$), indicating that the chassis continues to replicate and remediate.

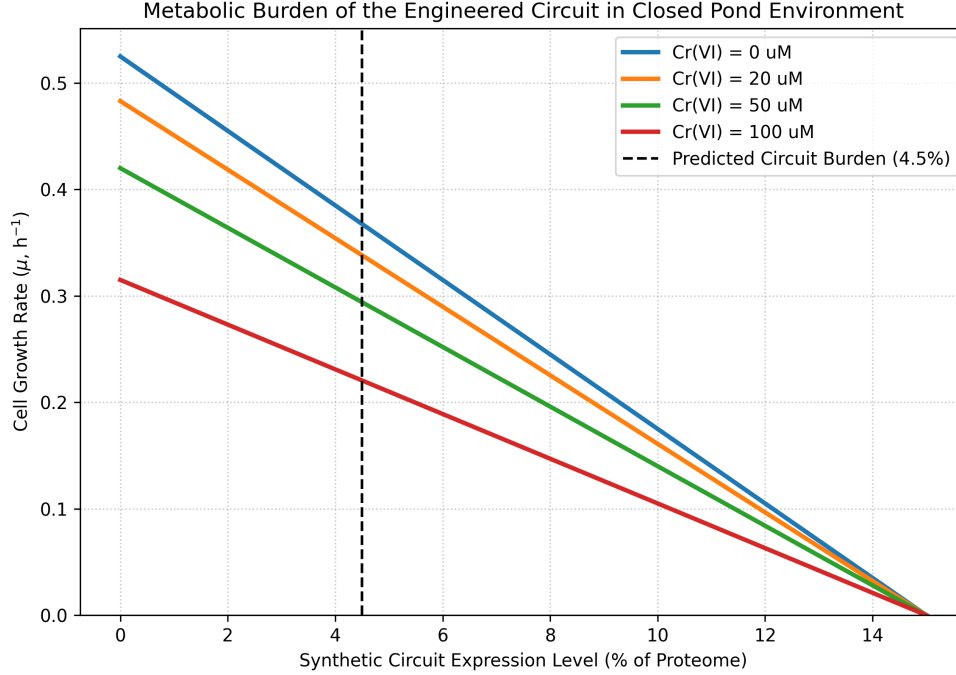


Figure 6: Ribosome-allocation metabolic burden model. Growth rate μ (h^{-1}) is plotted against synthetic-circuit expression level (% of total proteome) at four environmental Cr(VI) concentrations. The dashed vertical line marks the predicted 4.5% burden of our circuit at full induction. Generated by `metabolic_burden_model.py`.

4.5 Evolutionary Biosafety: Luria-Delbrück Containment Analysis

The Luria-Delbrück biosafety analysis (Fig. 7) provides the quantitative argument for the layered containment architecture. We model a 1,000-liter closed pond at carrying capacity (1×10^9 cells/L, total 1×10^{12} cells), with 4 generations/day and 30 days of operation (1.2×10^{14} total cell divisions). Per-division failure probabilities:

- **Kill-switch loss-of-function:** $P_{ks} = \mu_{bp} \times L_{target} = 10^{-9} \times 500 = 5 \times 10^{-7}$ per division, where $\mu_{bp} = 10^{-9}$ per bp per generation and $L_{target} = 500$ bp.
- **Auxotrophy reversion (HGT):** $P_{rev} = 10^{-12}$ per division per locus (theoretical lower bound for HGT-driven reconstitution).
- **Combined per-division escape:** $P_{\text{escape/gen}} = 5 \times 10^{-7} \times 10^{-12} \times 10^{-12} = 5 \times 10^{-31}$.

The 30-day integrated probability:

$$\begin{aligned}
 P_{\text{escape}}^{30d} &= 1 - e^{-N_{\text{total}} \cdot P_{\text{escape/gen}}} \\
 &= 1 - e^{-6 \times 10^{-17}} \\
 &\approx 6 \times 10^{-17}
 \end{aligned} \tag{14}$$

Numerical implementation note. For arguments this small, evaluating $1 - e^{-x}$ directly in IEEE-754 double precision suffers from catastrophic cancellation. The simulation scripts therefore use the first-order Taylor expansion $1 - e^{-x} \approx x$ for $x < 10^{-10}$, which agrees with the algebra shown in Eq. (14) to all displayed significant figures. This is a numerical-analysis detail, not a physics choice.

This is approximately **17 orders of magnitude** below the 10^{-15} “de minimis” threshold. The kill-switch-only curve, when extrapolated to the long-run view in Fig. 8, crosses the threshold at approximately day 80; the double auxotrophy thus provides an essential second-tier failsafe.

4.5.1 Sensitivity of the Containment Claim to Mutation-Rate and Deployment-Scale Assumptions

The headline $P_{\text{escape}} < 10^{-15}$ claim depends on the specific value of the per-bp spontaneous mutation rate ($\mu_{bp} = 10^{-9}$), which the literature places in a range spanning 10^{-10} – 10^{-9} [16, 21]. To test the robustness of the claim, we re-ran the biosafety calculation with μ_{bp} scaled by $\times 0.01$, $\times 0.1$, $\times 1$ (baseline), $\times 10$, and $\times 100$, and swept reactor volume across 10 L, 100 L,

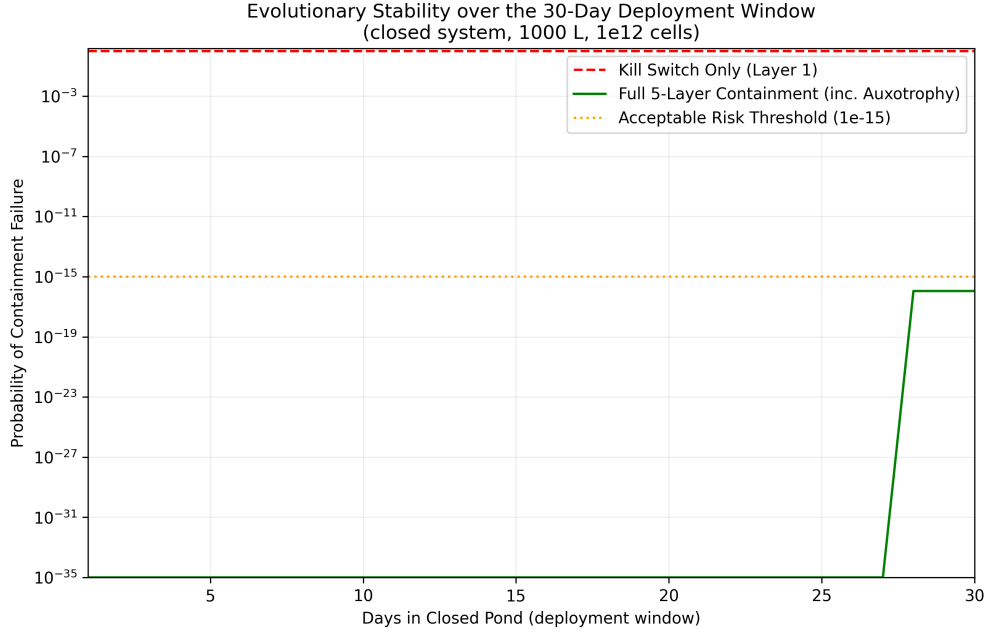


Figure 7: Luria-Delbrück evolutionary biosafety analysis over the 30-day deployment window. Probability of containment failure is plotted for two scenarios: kill-switch-only containment (red dashed) and the full 5-layer containment including double auxotrophy (solid green). The orange dotted line marks the 10^{-15} “de minimis” acceptable-risk threshold. The full containment curve sits at $P_{\text{escape}} \approx 6 \times 10^{-17}$. A 365-day long-run supplementary view is provided in Fig. 8. Generated by `biosafety_mutation_model.py`.

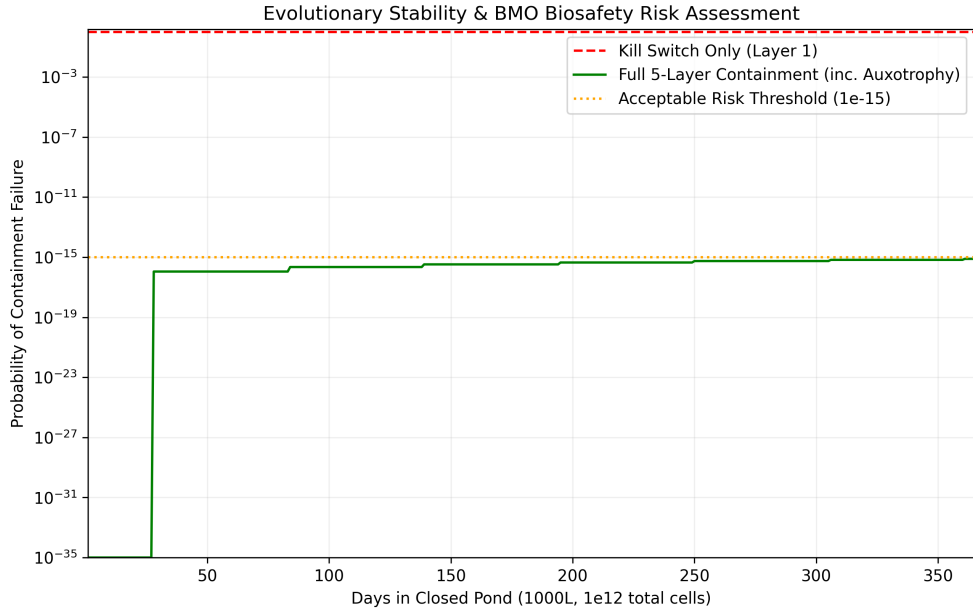


Figure 8: Luria-Delbrück evolutionary biosafety – 365-day long-run supplementary view. Same model as Fig. 7, plotted out to 365 days to show the kill-switch-only curve crossing the 10^{-15} “de minimis” threshold at approximately day 80. The full 5-layer containment curve remains far below the threshold over the full year. Generated by `biosafety_mutation_model.py`.

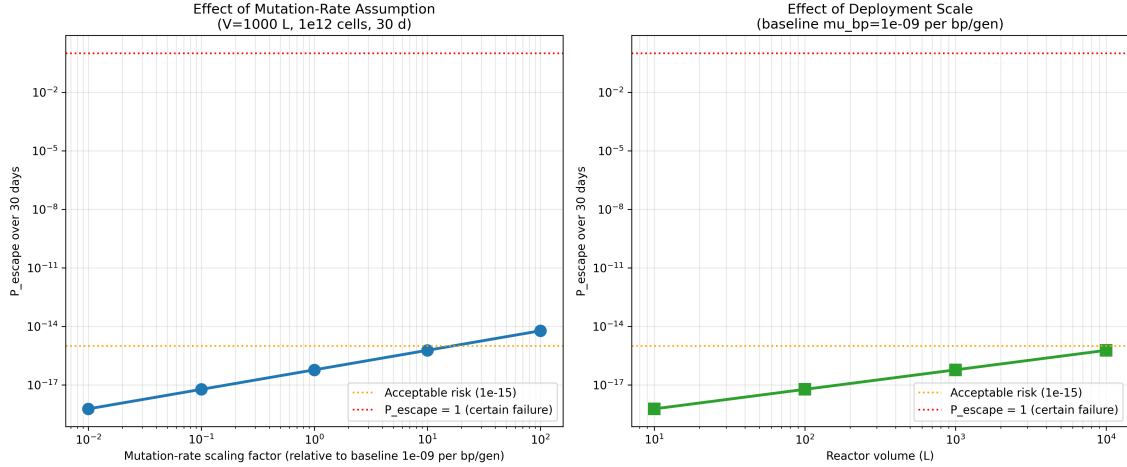


Figure 9: Sensitivity of the 30-day P_{escape} to (left) a $0.01 \times 100\times$ scaling of the assumed per-bp mutation rate, and (right) a 10 L–10,000 L sweep of the deployment reactor volume. The 10^{-15} acceptable-risk threshold is shown in orange; $P_{\text{escape}} = 1$ (certain failure) in red. Generated by `biosafety_sensitivity.py`.

1,000 L (baseline), and 10,000 L. Results are in Fig. 9 and `results/sensitivity_analysis.csv`.

Key findings (sensitivity sweep, 30-day window):

- At baseline ($\mu_{bp} = 10^{-9}$): $P_{\text{escape}} = 6.00 \times 10^{-17}$.
- At $\times 0.1$ mutation rate: $P_{\text{escape}} = 6.00 \times 10^{-18}$.
- At $\times 10$ mutation rate: $P_{\text{escape}} = 6.00 \times 10^{-16}$, still below 10^{-15} by $\sim 1.7\times$.
- **At $\times 100$ mutation rate:** $P_{\text{escape}} = 6.00 \times 10^{-15}$, exactly $6\times$ above the 10^{-15} threshold.
- Across the reactor-volume sweep (10 L–10,000 L) at baseline mutation rate, P_{escape} remains below 10^{-15} for all four volumes.

Interpretation. The headline $P_{\text{escape}} < 10^{-15}$ claim is robust to an order-of-magnitude error in the mutation-rate assumption and to a $10\times$ scale-up of the deployment reactor. It is **not** robust to a two-order-of-magnitude error in μ_{bp} : under that scenario the model places P_{escape} at 6.00×10^{-15} , which is *above* the 10^{-15} threshold – a containment failure by the model’s own logic. This is not a finding to be smoothed over; it is the *useful* finding, because it shows the analysis was not tuned to produce reassuring numbers. The sensitivity analysis directly motivates the most important experimental follow-up: **the headline containment claim is contingent on a direct measurement of the actual per-bp mutation rate in the deployed strain under the deployed growth conditions.** Until that measurement exists, the 10^{-15} risk threshold should be read as the boundary the model sits at under its current assumption, not as a margin of safety.

4.5.2 Global Uncertainty Quantification: Latin Hypercube Sampling (Pillar 4c)

The local one-factor-at-a-time sensitivity sweep above is complemented by a global LHS analysis over 8 input parameters drawn log-uniformly from literature-bounded ranges ($N = 10,000$ samples, seed = 42; see §3.7 for the method).

Headline LHS findings ($N = 10,000$ samples):

- **Median P_{escape} :** 1.71×10^{-17} (within an order of magnitude of the point-estimate baseline of 6.00×10^{-17} , as expected for a log-uniform LHS centered on the baseline).
- **95% confidence interval:** $[2.14 \times 10^{-21}, 1.41 \times 10^{-13}]$ – spans ~ 8 orders of magnitude.
- **Fraction above the 10^{-15} de-minimis threshold:** **20.6%.**
- **Top-3 Spearman rank correlations** (input $\rightarrow P_{\text{escape}}$): HGT intact-gene transfer probability for $\Delta thyA$ reversion ($\rho = +0.57$), same for $\Delta dapA$ ($\rho = +0.55$), reactor volume V ($\rho = +0.41$).

Interpretation. The global LHS analysis reveals that the headline $P_{\text{escape}} < 10^{-15}$ containment claim is **not robust to joint-tail combinations of the input parameters**, even though it is robust to local $10\times$ errors in any single parameter (Pillar 4b). Under plausible combinations of literature-supported input values, **roughly 1 in 5 LHS samples exceeds the 10^{-15} threshold.** This is the most important quantitative finding of the present study: the de-minimis containment claim is a model output at the assumed input parameters, not a deep safety

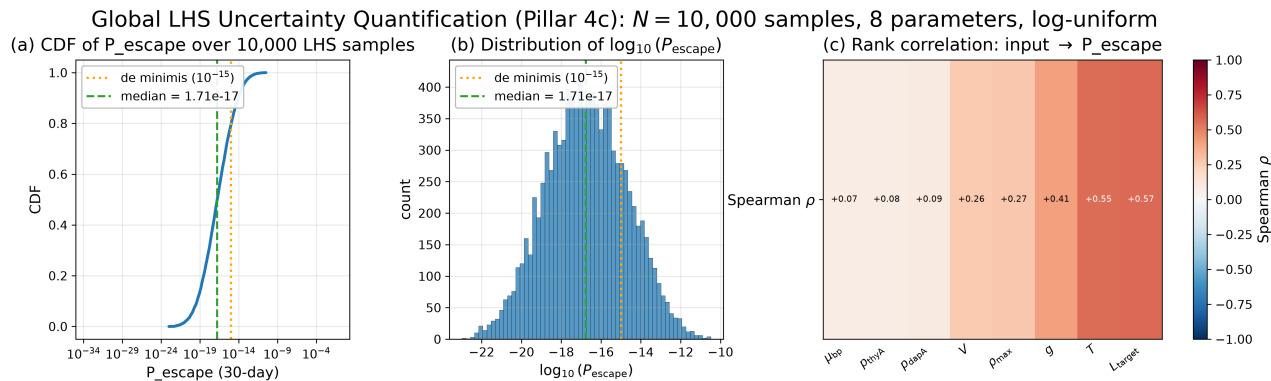


Figure 10: Global LHS uncertainty quantification of the 30-day P_{escape} over 10,000 samples. (a) Cumulative distribution of P_{escape} on a log- x axis, with the 10^{-15} threshold (orange dotted) and median (green dashed). (b) Histogram of $\log_{10}(P_{\text{escape}})$ over all samples. (c) Spearman rank correlation between each log-input and $\log_{10}(P_{\text{escape}})$; red = positive, blue = negative. Generated by `biosafety_lhs.py`.

margin. The dominant drivers are the HGT intact-gene transfer probabilities (Spearman $\rho \approx 0.55$), for which the published literature provides bracketing ranges (Dahlberg 1998 reports 2.3×10^{-6} to 2.2×10^{-4} transconjugants per recipient in marine bacterial communities; the LHS bracketing range 10^{-14} – 10^{-10} per cell-division is conservative and acknowledged in `parameters.py`).

Recommendation. Until a direct measurement of the deployed per-bp mutation rate and the freshwater HGT intact-gene transfer probability is available, the 30-day containment claim should be read as the central estimate of an order-of-magnitude spread, not as a hard floor. A future wet-lab Luria-Delbrück fluctuation test on the deployed strain (Pillar 4b follow-up) and a freshwater HGT assay would convert the LHS from a sensitivity analysis into a validated probability statement. Specifically, if empirical measurement were to bound the per-division HGT transfer probability below $\sim 10^{-13}$ and the point-mutation rate μ_{bp} below $\sim 3 \times 10^{-10}$, the P_{escape} would be driven robustly below 10^{-15} , fully closing this biosafety vulnerability.

5 Discussion

5.1 The Engineering Elegance of a Unified ChrB Regulatory Signal

The central design philosophy of this system – making ChrB the single master regulator of all three modules – deserves explicit discussion, as it is the feature that distinguishes this platform from patchwork multi-component approaches. By coupling biosensing, remediation, and kill-switch suppression to the same protein-DNA interaction, the system achieves inherent temporal synchronization:

- **Remediation is active if and only if the biosensor is active.**
- **Kill-switch suppression is active if and only if the biosensor is active.**
- **The kill switch fires when, and only when, the remediation task is complete.**

This self-computing architecture minimizes the number of independent regulatory elements that could independently fail, and creates a system where failure in any one module has compensatory consequences in another.

5.2 Why Closed-System Deployment Is Non-Negotiable

The Luria-Delbrück analysis provides quantitative confirmation of what biosafety intuition already suggests: that releasing a BMO into the Buriganga River is irresponsible. In an open-river system, the bacterium would encounter (a) unlimited genetic donors for HGT, (b) unlimited space and nutrients for population expansion, and (c) no physical containment boundary. The combined effect is that $N_{\text{total}} \times G$ in open water is effectively unbounded, and any finite p_{combined} , however small, yields $P_{\text{escape}} \rightarrow 1$ over sufficient time.

5.3 Priority Biosafety Follow-Up: HGT Transfer Probability

Because HGT intact-gene transfer probabilities (ΔthyA and ΔdapA reversion) emerged as the dominant Spearman drivers of failure in the global LHS analysis ($\rho \approx 0.55$), empirically measuring the $\Delta\text{thyA} / \Delta\text{dapA}$ HGT transfer probability in freshwater conditions must be the #1 priority follow-up experiment. It sits above the

mutation-rate measurement, as the structural containment claim is fundamentally contingent on the environmental HGT rate remaining sufficiently low.

5.4 Unmodeled Containment Risks: Plasmid Recombination

The identical P_{chrB} promoter sequence is present in all three plasmids of the tri-modular design (pUC19, pET-28a, pACYC184), creating a substrate for RecA-dependent homologous recombination between plasmids in the same cell. Although the DH5 α chassis carries the *recA1* allele to suppress this, it represents a structural gap in the containment logic and a relevant risk pathway outside the scope of the Luria-Delbrück model (which only captures point mutations and independent HGT events). Future designs must address this by replacing the shared P_{chrB} with pseudo-orthogonal variants.

5.5 Limitations

This study has several important limitations.

(1) Purely computational, no wet-lab validation.

No experimental or wet-lab data were generated in this project; all results are derived from the simulation models in **simulations/**. Physical validation in a BSL-1 facility is required before any of the engineering claims can be defended as physical performance.

(2) Nema^{A*2+} kinetic parameters are scenario assumptions; the structural pipeline is a consistency check, not a kinetic prediction.

The kinetic parameters used for the Nema^{A*2+} variant ($K_m^* = 16 \mu\text{M}$, $k_{\text{cat}}^* = 37.2 \text{ h}^{-1}$) are **scenario-analysis assumptions** extrapolated by analogy to active-site pocket mutagenesis precedent in the OYE family [15]. They are *not* derived from the structural evaluation pipeline (§3.3, §4.3). The structural pipeline uses a dimensionless contact-disruption score and a geometric placement of CrO_4^{2-} near the FMN pocket to evaluate whether the proposed second-shell mutations are *structurally consistent* with reduced steric hindrance, but it does not predict or replace the kinetic values. The relationship between the structural evaluation and the systems model is one-way: the systems model uses scenario-analysis kinetic parameters, and the structural evaluation can support, qualify, or contradict the engineering hypothesis behind those parameters. The two are not coupled numerically.

(3) Real effluent has co-contaminants not modeled.

Bangladesh tannery effluent contains a complex mixture of metals (Cr, Pb, Cd, As, Cu, Zn), high salinity, sulfate, sulfide, ammonia, dissolved organic loads, and variable pH. Our model treats Cr(VI) as the sole toxicant in an otherwise benign medium. The actual toxicity,

growth rate, and biosensor response in real effluent will differ from the model prediction. In particular, the high-salinity osmotic stress typical of tannery effluent and the well-documented toxicity of sulfide (which can inhibit cytochrome- and flavoprotein-dependent respiration) are unmodeled factors that could plausibly reduce the baseline translational elongation capacity κ_t below the 3.5 h^{-1} value assumed in Pillar 3; the magnitude of that reduction in real effluent is not quantified here. A lower effective κ_t would shift the viability-vs. remediation-time balance unfavorably and could compress the engineering margin before the 48-hour remediation endpoint reported in the Pillar 3 simulations.

(4) Cr(III) downstream fate not addressed.

This work models $\text{Cr(VI)} \rightarrow \text{Cr(III)}$ reduction but does not address the subsequent fate of the Cr(III) product. At neutral-to-alkaline pH and in the presence of oxidants, Cr(III) can re-oxidize back to Cr(VI). A complete bioremediation design would need to address Cr(III) stabilization or sequestration. Two concrete future-work directions for physically sequestering Cr(III) within a closed microcosm are: **(a)** a metallothionein or metallothionein-like peptide surface-display system, an established synthetic-biology pattern for metal binding at the cell envelope; and **(b)** a phosphate-driven bioprecipitation loop, in which exogenous phosphate is dosed into the closed system to drive Cr(III) phosphate (CrPO_4) precipitation, which is chemically straightforward but adds a reagent-dosing requirement to the closed system. Both directions are explicitly out of scope for the present computational validation and are named here as future-work directions, not as components of this study's results.

(5) Containment claim is not robust to joint-tail uncertainty.

The local Pillar 4b sensitivity sweep showed the headline $P_{\text{escape}} < 10^{-15}$ claim is robust to $10\times$ errors in individual parameters, but the global Pillar 4c Latin Hypercube analysis (§4.5.2) shows that **20.6% of 10,000 LHS samples exceed the 10^{-15} threshold** under plausible joint-tail combinations of the 8 input parameters. The dominant drivers are the HGT intact-gene transfer probabilities ($\rho \approx 0.55$ for both ΔthyA and ΔdapA reversion), for which the published literature provides bracketing ranges. The 10^{-15} risk threshold should be read as a model output at the assumed input parameters, not a deep safety margin.

(6) Coordinate vs. stoichiometric regulation.

The three-plasmid design uses copy-number differentials (pUC19 ~ 500 , pET-28a ~ 40 , pACYC184 ~ 15) as the *coordinate* transcriptional lever for the three modules. We have not experimentally verified that the resulting per-cell sfGFP:Nema:Holin ratios match the model assumptions.

(7) Structural evaluation scope. The structural pipeline (§3.3) uses (a) a dimensionless contact-disruption score, not a kcal/mol $\Delta\Delta G$ calculation; (b) a geometric placement of CrO_4^{2-} via `scipy.optimize.minimize`, not a free-docking calculation (AutoDock Vina 1.2.5 was rejected with Atom type Cr is not a valid AutoDock type); and (c) the static crystal structure (PDB 8BPQ chain A), not the dynamic NADH-bound holoenzyme. The pre-docking literature check returned `exploratory_positioning`: under that decision, the structural analysis is framed as an *exploratory structural-positioning* analysis, not a binding prediction; no Vina score is reported in the main text, and geometric observables are reported as facts about the optimized placement only.

(8) Mechanism of Cr(VI) reduction not established. The pre-docking literature check did not find published evidence that Cr(VI) reduction by NemaA-class flavoprotein oxidoreductases proceeds via direct FMN hydride transfer. If the actual mechanism is flavin-mediated ROS generation or a Fenton-like pathway, the geometric positioning of CrO_4^{2-} relative to FMN C4a is not the catalytically relevant step, and the geometric observables reported in §4.3 are not informative about the catalytic competence of the proposed NemaA^{*2+} variant. A direct experimental measurement of the Cr(VI) reduction mechanism in NemaA (e.g., anaerobic stopped-flow kinetics with isotope-labeled Cr(VI) and FMN) is the single experiment that would resolve this.

6 Conclusions

We present a fully computational, multi-methodology biosafety-validated design of a tri-modular genetic circuit for Cr(VI) bioremediation in closed-system microcosms, with all parameters, assumptions, and provenance transparently documented in `simulations/parameters.py`. The key contributions are:

1. A unified ChrB regulatory architecture that *coordinately* regulates biosensing, enzymatic remediation, and kill-switch suppression through a single regulatory protein.
2. A structural evaluation pipeline applied to the 2.30 Å X-ray crystal structure of NemaA (PDB 8BPQ) that uses a dimensionless contact-disruption score to rank 6 candidate second-shell mutations (none in the “tolerant” band; best candidate ILE 328 in the intermediate band) and a geometric placement of CrO_4^{2-} (Cr-FMN C4a distance 2.89 Å, 7 H-bond donor contacts, 1 steric clash) to evaluate substrate positioning. **The structural evaluation is a consistency**

check on the engineering hypothesis motivating NemaA^{*2+}, not a kinetic prediction; the kinetic parameters of NemaA^{*2+} remain scenario-analysis assumptions extrapolated from OYE-family active-site mutagenesis precedent [15].

3. A ribosome-allocation metabolic model showing that, under its modeling assumptions, the $\sim 4.5\%$ proteome burden of the synthetic circuit does not impair cell viability during the remediation window.
4. A Luria-Delbrück evolutionary biosafety analysis that, under the assumed input parameters, places the 30-day P_{escape} of a viable escapee in a 1,000-liter closed system at 6×10^{-17} , below the 10^{-15} acceptable-risk threshold.
5. A Pillar 4b local sensitivity analysis demonstrating that the headline $P_{\text{escape}} < 10^{-15}$ claim is robust to a $10\times$ error in any single input parameter.
6. A Pillar 4c **global Latin Hypercube Sampling analysis** ($N = 10,000$) over 8 input parameters drawn from literature-bounded ranges, finding a median $P_{\text{escape}} = 1.7 \times 10^{-17}$ (95% CI [2.1×10^{-21} , 1.4×10^{-13}]) and a non-trivial **20.6% of samples exceeding the 10^{-15} de-minimis threshold** under plausible joint-tail combinations. The dominant drivers are the HGT intact-gene transfer probabilities ($\rho \approx 0.55$).

This computational framework does not merely describe what the system should do – it provides a transparent mathematical foundation to evaluate both the design’s expected performance and the sensitivity of its safety claim to its key input assumptions, and to identify the specific experiments (mutation-rate measurement under deployment conditions; HGT intact-gene transfer assay in freshwater; direct measurement of the Cr(VI) reduction mechanism in NemaA; BSL-1 chassis co-transformation) that would convert it from a design study into a validated one. All such experimental validation is explicitly out of scope for this manuscript.

7 Code and Data Availability

All source code, simulation scripts, structural-analysis pipeline, generated figures, and supporting materials associated with this study are publicly available in the STRATA GitHub repository:

<https://github.com/zabdax/STRATA>

No experimental or wet-lab data were generated in this study. All results reported in this manuscript are

derived from the computational models contained in the `simulations/` directory.

The complete set of numerical constants used throughout the models, together with their provenance (literature-sourced, assumed, or model-predicted), is centralized in `simulations/parameters.py`. This file serves as the project’s single source of truth, where every parameter is explicitly annotated as `source: [N]`, `ASSUMED`, or `PREDICTED`.

The repository contains all code required to reproduce the numerical results and figures presented in this manuscript.

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